Fiscal Year:	FY 2005	**Task Last Updated:**	FY 11/12/2004
PI Name:	Colver, Gerald Ph.D.		
Project Title:	Quenching of Particle-Gas Combustible Mixtures Using Electric Particulate Suspension (EPS) and Dispersion Methods		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	gmc@iastate.edu	**Fax:**	FY 515-294-3261
PI Organization Type:	UNIVERSITY	**Phone:**	515-294-1423
Organization Name:	Iowa State University		
PI Address 1:	2025 Black Engineering Building, Department of Mechanical Eng		
PI Address 2:			
PI Web Page:			
City:	Ames	**State:**	IA
Zip Code:	50011-2161	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	99-HEDS-04
Start Date:	01/09/2001	**End Date:**	12/07/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	5
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
Contact Monitor:	Yuan, Zeng-Guang	**Contact Phone:**	216-433-2838
Contact Email:	zengguang.yuan-1@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Goroshin, Samuel (McGill University) Lee, John (McGill University)		
Grant/Contract No.:	NCC3-846		
Performance Goal No.:			
Performance Goal Text:			
	The accurate measurement of flame-quenching distance for dusts is an important parameter in fire safety applications and also in advancing laminar fire flame theory as it is closely related to the characteristic flame thickness and structure. Two suspension techniques are being used to evaluate powder quenching distance in combustible mixtures: ISU's electric particle suspension (EPS) method and McGill's flow dispersion method. Quenching by inert particles is also possible. Tests carried out in microgravity will permit the formation of higher and more uniform dust concentrations with higher concentrations and larger particles compared to standard gravity using EPS. Both gravity environments can be studied with EPS. Particle velocity distribution has heretofore not been investigated as a controlled variable in powder combustion systems nor has relative velocity of particles with oxidizer. Measurements will support theoretical modeling of flame propagation in dust suspensions. Microgravity tests will be conducted in the 2.2-second drop tower and on the KC-135. The objectives of this research are to experimentally measure flame quenching distance,		

Task Description:	flammability limits, and burning velocities in uniform particulate suspensions of solids over a wide range of dust concentrations and particle sizes in microgravity. A new found objective is to package the EPS into a compact “suitcase” design suitable for ignition testing of powders in space and in the field. A final objective is to determine the effect of particle slip velocity and velocity distribution on flame propagation. The EPS method has potential as a 'benchmark' design for one-dimensional flame propagation in dusty gas mixtures thereby providing NASA with a fire safety standard. Achieving a uniform dust concentration is now possible and has long been identified as a key problem to be resolved in dust flame studies. Possible uses of this new EPS technology includes space (microgravity) research and field testing of combustible powders for fire safety and for use in Martian atmospheres containing CO2 and magnesium. Flame quenching distance of dusts is also an important parameter in laminar flame theory as it is closely related to the characteristic flame thickness and structure. The EPS method also permits the characterization of powder combustion propagation with particle velocity. Reliable measurements of flame quenching distance, particle velocity, and flammability limits will serve as a basis for theoretical modeling of the mechanisms for flame propagation in dust suspensions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	ISU's Electric Particulate Suspension (EPS) is a unique suspension technique highly suitable for carrying out powder combustion studies in normal gravity. A compact “suitcase” EPS design is also possible as portable fire safety instrument. The present fundamental research will be of particular interest to those working in laboratories and industry across the United States, particularly those concerned with high technology combustion research and also safe handling of powders.
Task Progress:	A specially designed 5-tiered rig for taking multiple measurements of ignition and quenching distance of combustible powder-gas mixtures in normal gravity and microgravity has been taken from concept to working prototype. The combustion rig was designed utilizing computer software and tested to withstand 30 g’s for use in the 2.2 second NASA Glenn drop tower. The design permits up to 5 simultaneous data points to be tested (per drop) for better defined quenching curves (fuel-oxidant ratio with quenching distance). Technical developments completed and tested include the following: • Application of in situ particle (instantaneous) photo diode detection system for concentration measurement using an optically expanded single laser beam. Multiple point “instantaneous” measurements are important in evaluating cloud stratification (cloud uniformity) during drop tower tests which serves to upgrade our previous single beam scan method used in ground tests. • Use of two 25 kV high voltage sources for independent control of suspension and ignition of powders– this supports an early goal permitting the relative velocity effect of particles and oxidant to be studied (an improvement over early studies in quenching). • A special high voltage capacitor circuit with computer controlled spark trigger relays that delivers variable spark energy to 12.5 J per module for igniting powders. • A unique shaker table test rig (to 15 g’s) was designed, fabricated, and built. Vibration also aids suspension of cohesive fine powders. Tests showed no EPS cell damage or glass breakage.
Bibliography Type:	Description: (Last Updated: 06/01/2010)
Abstracts for Journals and Proceedings	Colver GM, Greene N, Xu H. "EPS (Electric Particulate Suspensions) Microgravity Technology Provides NASA with New Tools." Strategic Research to Enable NASA’s Exploration Missions Conference, Cleveland Marriott, Cleveland, OH, June 22-23, 2004. Proceedings, Strategic Research to Enable NASA’s Exploration Missions Conference, Cleveland, OH, June 22-23, 2004. p. 65 , Jun-2004
Abstracts for Journals and Proceedings	Greene NJ, Shoemaker DM, Xu H, Colver GM. "Instrumentation and Design Considerations for a New Electrostatic Powder Combustion Testing Method." 115th Annual Iowa Academy of Science, Hotel Fort Des Moines, Des Moines, Iowa, Apr. 25-26, 2003. 115th Annual Iowa Academy of Science, Hotel Fort Des Moines, Des Moines, Iowa, Apr. 25-26, 2003. , Apr-2003
Articles in Peer-reviewed Journals	Colver GM, Greene N, Shoemaker D, Kim S, Yu T. "Quenching Dust Mixtures: A New Microgravity Testing Method Using Electric Particulate Suspensions." AIAA Journal 2004 Oct;42(10):2092-100. , Oct-2004
Dissertations and Theses	Greene N. "Cloud Stratification Measurement of an Electric Particulate Suspension for Studies in Gravity and Microgravity." MS Thesis, Dept. of Mechanical Eng., Iowa State University, Ames, IA, November 2004. , Nov-2004
NASA Technical Documents	Colver GM, Greene N, Shoemaker D, Xu H. "Quenching Combustible Dust Mixtures Using Electric Particulate Suspensions (Eps): A New Testing Method For Microgravity." Seventh International Workshop on Microgravity Combustion and Chemically Reacting Systems, The Physical Sciences Div of NASA Office of Biological and Physical Research (OBPR), Cleveland, OH, June 3-6, 2003. NASA/CP-2003-212376, p. 173-176. , Jun-2003
Papers from Meeting Proceedings	Colver GM, Eimers C, Greene N. "Quenching of Combustible Dust Mixtures Using Electric Particulate Suspension (EPS): Review of a New Testing Method for Microgravity." 41st AIAA Aerospace Sciences Meeting and Exhibit, Reno, NV, January 6-9, 2003. 41st AIAA Aerospace Sciences Meeting and Exhibit, Reno, NV, January 2003. Paper AIAA 2003-705. , Jan-2003
Presentation	Greene, N. J.; Shoemaker, D. M.; Xu, H., and Colver, G. M. "Instrumentation and Design Considerations for a New Electrostatic Powder Combustion Testing Method" 11 th Annual Iowa Academy of Science, Hotel Fort Des Moines, IA. Apr-2003